MAHALAKSHMI

Mail | Linkdein | github | portfolio

PROFESSIONAL SUMMARY

AI and Data Science fresher skilled in Python, SQL, Machine Learning, Deep Learning, and Generative AI. Experienced in building end-to-end ML pipelines — data cleaning, EDA, feature engineering, model training, hyperparameter tuning, and Streamlit deployment. Proficient in Predictive Analytics, Data Visualization, NLP, and Computer Vision. Strong problem-solving and analytical thinking with hands-on projects spanning customer analytics, revenue forecasting, and AI-driven applications. Familiar with Git, GitHub, VS Code, and PyCharm.

PROJECTS

Customer Retention Intelligence System *| Python, Pandas, NumPy, Scikit-Learn, XGBoost, Streamlit*

- Built end-to-end ML pipeline for customer churn prediction; performed data cleaning, EDA, and feature engineering on customer analytics datasets to surface retention-critical patterns.
- Trained and compared XGBoost, Random Forest, and Logistic Regression with hyperparameter tuning; evaluated using accuracy, precision, recall, and F1-score; deployed via interactive Streamlit dashboard.

Business Revenue Forecasting System *| Python, Pandas, NumPy, Scikit-Learn, Matplotlib, Streamlit*

- Developed a time series regression forecasting system to predict future business revenue; conducted EDA and feature engineering on historical data to identify seasonal trends and key predictors.
- Built Streamlit dashboards with Matplotlib visualizations to translate model outputs into actionable business intelligence for stakeholder decisionmaking.

Economic Recovery Intelligence System *| Python, Pandas, NumPy, Scikit-Learn, XGBoost, SHAP, Streamlit*

- Engineered ML pipeline using macroeconomic indicators (GDP, inflation, unemployment) for economic recovery prediction; applied SHAP for model interpretability and reported F1-score and accuracy benchmarks.
- Deployed real-time Streamlit dashboard for interactive visualization and scenario-based economic prediction.

Mental Health Detection using Deep Learning *| Python, TensorFlow, Keras, NLP, Streamlit*

- Built NLP-based deep learning classification system using text preprocessing, tokenization, and neural network architectures; improved recall and F1score via iterative hyperparameter tuning.
- Deployed Streamlit web app for real-time mental health assessment, demonstrating end-to-end AI deployment capability.

Emotion-Aware Generative AI Chatbot *| Python, CNN, TensorFlow, Gemini API, Streamlit*

- Trained CNN on facial emotion datasets for real-time emotion classification; integrated Gemini API to generate personalized responses, showcasing applied Generative AI and Computer Vision in a deployed Streamlit app.

CERTIFICATIONS

Essential SQL Skills for Data Beginners | FreeCourse | November 2025 | certificate

TECHNICAL SKILLS

Languages: Python, SQL
Libraries: Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, XGBoost, SHAP, Matplotlib, Seaborn
ML / DL: Classification, Regression, Predictive Analytics, Feature Engineering, EDA, Hyperparameter Tuning, NLP, CNN, Computer Vision
Generative AI: Gemini API, LLM Integration
Deployment & Viz: Streamlit, Matplotlib, Seaborn
Tools & DB: Git, GitHub, VS Code, PyCharm, MySQL
Concepts: ML Pipelines, Data Cleaning, Statistics, Business Intelligence, Analytical Thinking, Problem Solving, Critical Thinking

EDUCATION

B.A. Economics | Bharathidasan University, Thanjavur, Tamil Nadu | May 2025 | CGPA: 7.0/10